COSC3106: Theory of Computing

CH#2: Nondeterminism:

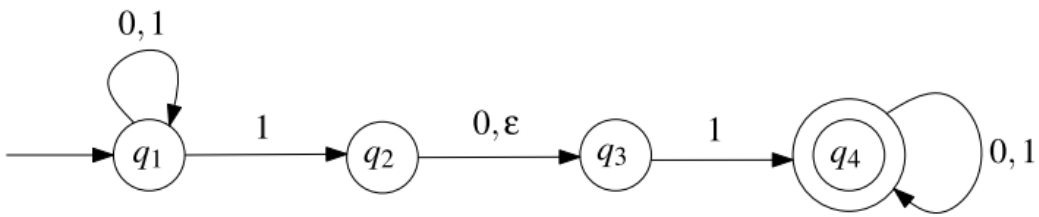
Lec#4: Non-deterministic Finite Automaton

School of Computer Science and Technology

This lecture

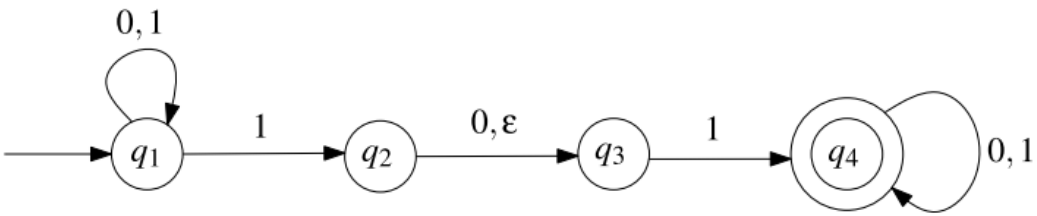
- By the end of this lecture, you will …
  - Understand the concept of the non-deterministic finite automaton

- Let us start by giving examples of nondeterministic finite automata…
  - Ex#1: Consider the following state diagram:



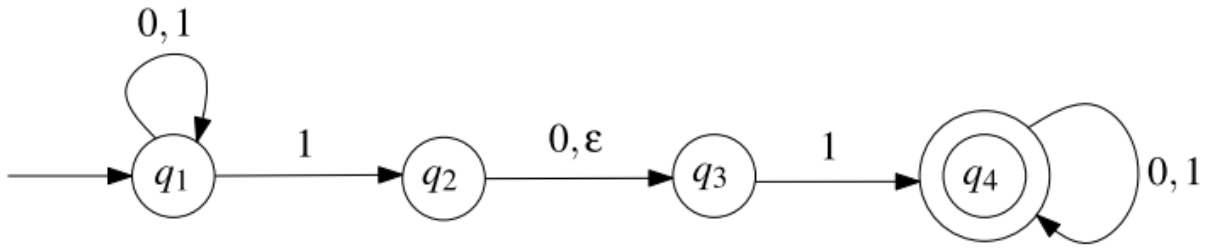
  - You will notice three differences with the FA that we have seen until now:
    1. If the automaton is in state q_1 and reads the symbol 1, then it has two options: Either it stays in state q_1, or it switches to state q_2 .
    2. If the automaton is in state q_2, then it can switch to state q_3 without reading an input symbol; this is indicated by the edge having the empty string ε as label.
    3. If the automaton is in state q_3 and reads the symbol 0, then it cannot continue.

- Ex#1 (cont.): What this automaton can do when it gets the string 010110 as input.



- An NFA accepts a string, if there exists at least one path in the state diagram that:
  - (i) starts in the start state,
  - (ii) does not stuck before the entire string has been read, and
  - (iii) ends in an accept state.

- **Ex#2: If the input string is 010**



- **In this case, there are three possible computations:**

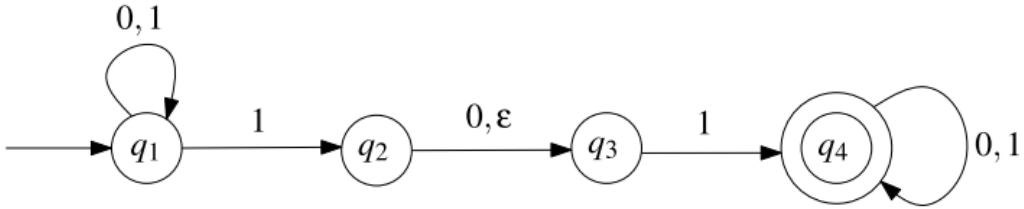
$$q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_1$$

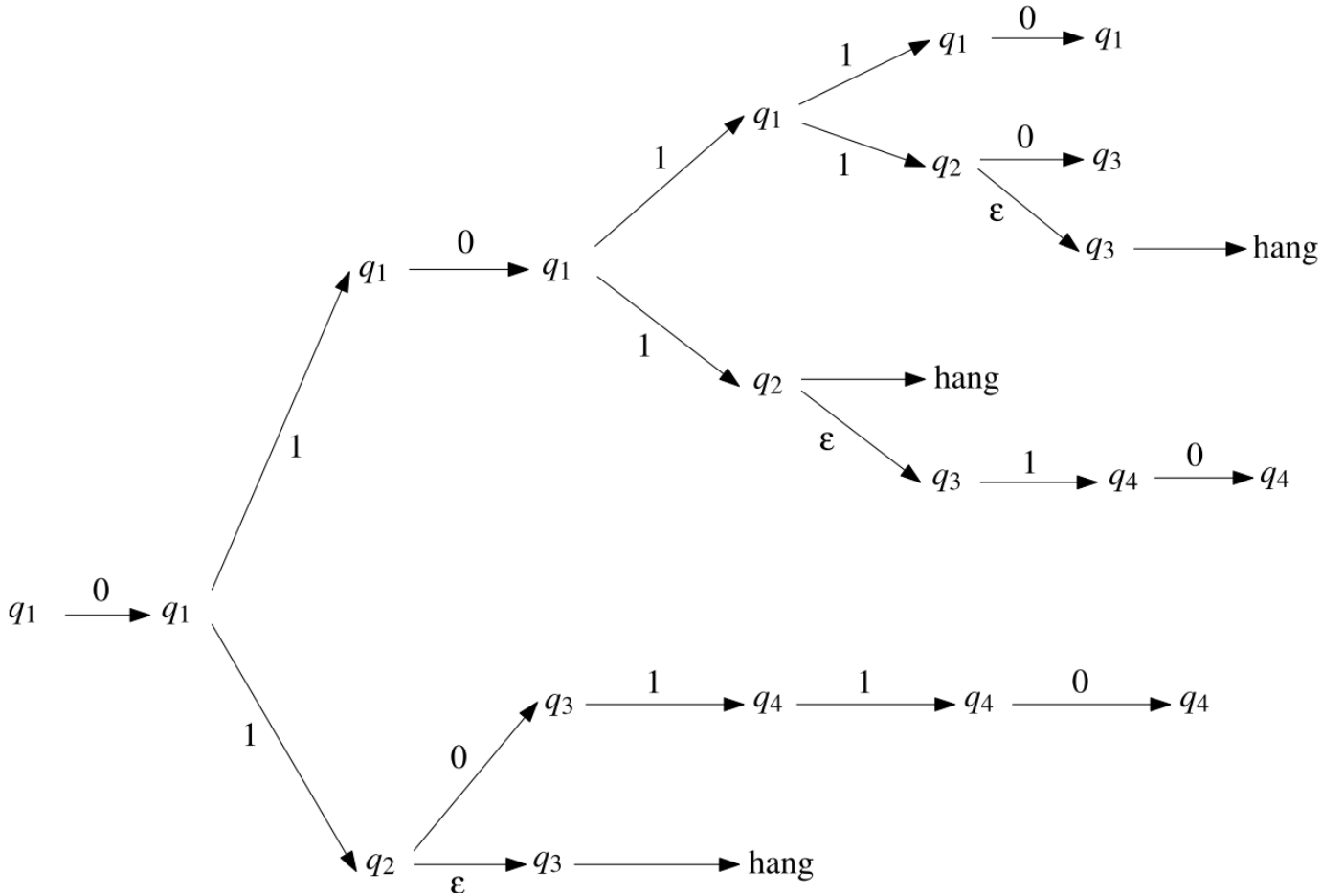
$$q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{0} q_3$$

$$q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \xrightarrow{\epsilon} q_3 \rightarrow \text{hang}$$

- **A string w is accepted by this NFA if and only if the input string contains 101, or 11**

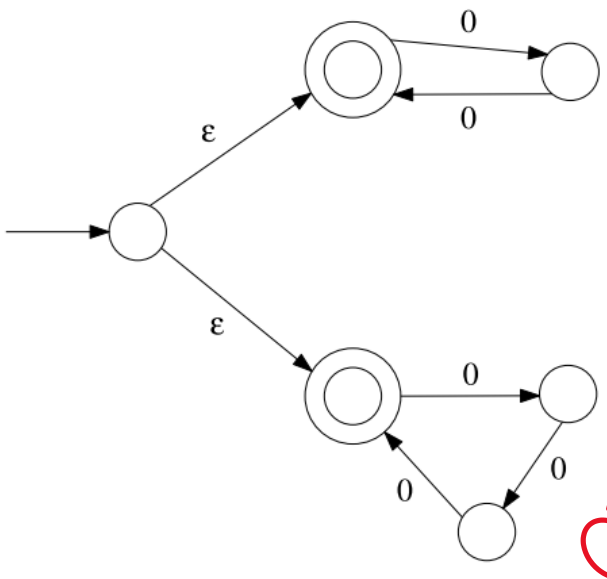
- **Ex#2: If the input string is 010110**





- **Ex#3: Let A be the language $A = \{ w \in \{0, 1\} : w$ has a 1 in the third position from the right $\}$.**
  - **We need to create an NFA that accepts A.**

- **Ex#4: Consider the following state diagram, which defines an NFA whose alphabet is $\{0\}$.**



0^k; k is even

- **Define its accepted language?**

$\{w\colon\ 0^k;\ k$ **is even or multiples of** $3\}$

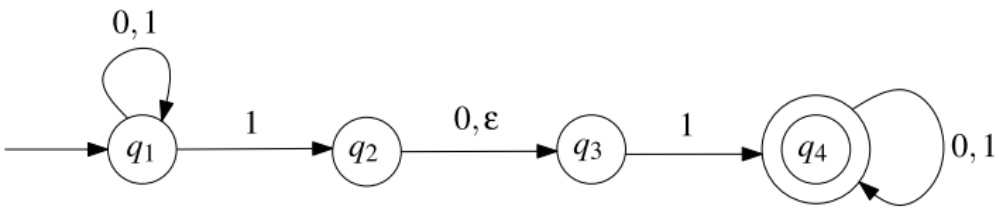
0^k ; k: multiples of 3

- Definition of nondeterministic finite automaton:
  - The previous examples give you an idea what nondeterministic finite automata are and how they work.
  - In this section, we give a formal definition of these automata.
    - For any alphabet Σ, we define Σ_ε to be the set: $\Sigma_\varepsilon = \Sigma \cup \{\varepsilon\}$

NFA: $N = \{Q, \Sigma, \delta, q, F\}$

- Q is a finite set, whose elements are called states,
- Σ is a finite set, called the alphabet; the elements of Σ are called symbols,
- $\delta : Q \times \Sigma_\varepsilon \to P(Q)$ is the transition function,
- q is the start state,
- F is a subset of Q; the elements of F are called accept states.

- Ex: Consider the NFA in the figure, specify it as a formal NFA?



- $Q = \{q_1, q_2, q_3, q_4\}$,
- $\Sigma = \{0, 1\}$,
- The start state is q_1,
- The set of accept states is $F = \{ q_4 \}$.
- The transition function δ is given by the following table:

	0	1	ϵ
q_1	$\{q_1\}$	$\{q_1, q_2\}$	$\emptyset$
q_2	$\{q_3\}$	$\emptyset$	$\{q_3\}$
q_3	$\emptyset$	$\{q_4\}$	$\emptyset$
q_4	$\{q_4\}$	$\{q_4\}$	$\emptyset$

- N accepts string $w \in \Sigma^*$
  - if $w = y_1 y_2 \dots y_m$; $y_i \in \Sigma_\varepsilon$
  - $\exists$ a sequence of states:

$$r_0, r_1, \dots, r_m \in Q$$

  - $r_0 = q$
  - $r_{i+1} \in \delta(r_i, y_{i+1}), i = 0, 1, 2, \dots, m-1$
  - $r_m = F$

- In Words, an NFA accepts a string, if there exists at least one path in the state diagram that:
  (i) starts in the start state,
  (ii) does not stuck before the entire string has been read, and
  (iii) ends in an accept state.

  - A string for which (i), (ii), and (iii) does not hold is rejected by the NFA.